Three-Timescale Nested Control of Residential EV Charging: Learned Pricing and Dispatch With Voltage Support From Charger Reactive Power

Abraheem Rashid, Faisal Iradat, Waseem Iqbal, and Yawar Abbas Bangash

***Abstract*—Residential electric-vehicle (EV) charging concentrates in the evening, when energy is expensive and feeder voltages are lowest. Prices change hourly, dispatch every few minutes and voltages within a minute, yet most learned coordinators act at one rate. We propose a nested controller with one level per timescale. Proximal policy optimization sets an hourly price corridor. A shared deep deterministic policy gradient agent dispatches every $15\,\mathrm{min}$ as a learned residual on a deadline-aware cheapest-slot plan, and least-laxity-first allocation keeps dispatch within charger and transformer limits. A non-learned layer corrects voltage every $60\,\mathrm{s}$ with charger reactive power bounded by the inverter rating. On IEEE 33- and 69-bus feeders with AC power flow, real prices and household loads, a synthetic residential fleet and, for generalization, recorded workplace sessions (50 held-out days), the controller costs 30.8% less than uncoordinated charging at 80% penetration without voltage violations and is the cheapest learned controller, but 4.1% more than a price-aware heuristic. Ablations attribute the cost gains to the planning prior and the absence of violations to the voltage layer. Attached to any baseline, it removes all violations up to 100% penetration whenever the base load alone respects the floor.**

***Index Terms*—Deep reinforcement learning, distribution networks, electric vehicle charging, multi-timescale control, reactive power, residual policy learning, transportation electrification, voltage regulation.**

I. Introduction

TRANSPORTATION electrification is accelerating: global electric-car sales exceeded 17 million in 2024 and are expected to pass 20 million in 2025 [1]. Electric vehicles (EVs) are thereby becoming one of the largest flexible loads on distribution feeders, and *when* they charge decides both what drivers pay and whether the feeder stays within its voltage limits. Residential charging concentrates in the evening, when household demand peaks. On weak radial feeders it depresses end-of-line voltages toward and below the $0.95\,\mathrm{p.u.}$ lower limit of the ANSI C84.1 service range [2], [3] and buys energy in the most expensive hours of the day-ahead market. Both effects are avoidable, because most vehicles stay plugged in far longer than they need to charge: a coordinator that shifts charging into cheap, lightly loaded hours lowers the energy bill and relieves the feeder at the same time.

The coordination problem addressed here is the following. A distribution system operator and a set of charging aggregators serve a residential fleet on a radial feeder. Each plugged-in vehicle has a known remaining energy need and departure time; the aggregators see day-ahead prices a limited number of hours ahead and measure their bus voltages; drivers accept or decline the retail price they are offered. The controller must choose retail prices and per-vehicle charging rates that minimize the day-ahead energy cost of the fleet, while every bus voltage stays at or above the floor, every charger and aggregator transformer stays within its rating, every vehicle leaves charged whenever its dwell allows it, and the mean retail price paid stays at the flat tariff that the drivers would otherwise pay (revenue neutrality). Four properties of this problem shape our design. First, its inputs change at different rates: prices change hourly, the plugged-in fleet and its needs change every few minutes, and bus voltages change within seconds, so a single decision interval fits none of them well. Second, voltage safety is a run-time requirement, not an average one; a controller that treats the floor as a reward penalty or an expected cost can still violate it on individual days. Third, feasibility of the dispatch (charger, transformer and deadline limits) is a hard constraint that penalties teach only approximately. Fourth, the value of any learned coordinator can only be established against strong model-based references, with the same voltage correction available to all, over many held-out days and several training seeds.

Prior work addresses these properties separately. Optimization-based schedulers [3], [4] need vehicle-level forecasts over the whole horizon and, to see the network, a network model. Decentralized schemes such as time-of-use (TOU) tariffs, game-theoretic demand response and priority rules [5]–[7] scale well but do not observe the network; serving vehicles with less laxity first [7] underlies our allocation. Deep reinforcement learning (RL) has scheduled real-time charging with deep Q-learning and price features [8] and with DDPG [9], exploited the structure of optimal policies in a distribution network [10], coordinated multiple agents [11], scaled with graph neural networks [12] and priced charging stations [13]; EV2Gym [14] and SustainGym [15] provide benchmarks and, here, the open data. Hierarchical designs coordinate several stations [16] or regulate voltage on two timescales for flash-charged buses [17]. Safe RL handles constraints such as voltage limits as expected costs

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TABLE I
POSITIONING BY METHOD CLASS (RL: REINFORCEMENT LEARNING; –: NOT CHARACTERIZED HERE)

Class	Rates	Voltage	Limits
Network optimization [3]	one	network model	constraints
Household scheduling [4]	one	not modeled	constraints
Decentralized [6], [7]	one	not modeled	per vehicle
Deep RL charging [8], [9]	one	not modeled	action bounds
Safe RL [18], [20]	one	expected cost	–
Two-timescale deep RL [17]	two	deep RL	–
Nested (ours)	three	measured, corrected	by construction

(CPO [18] and Lagrangian methods, reviewed in [19]) and has been applied to Volt-VAR control [20], active distribution networks [21], stability-constrained [22] and multi-agent [23] voltage control. EV chargers have supported voltage with vehicle-to-grid active power [24], reactive power [25] or both [26]; inverter reactive capability is standardized in IEEE Std 1547-2018 [27]. Table I summarizes the gap: no published coordinator acts at the three rates of the problem, enforces the voltage floor at run time from measurements, and keeps dispatch feasible by construction, while being compared with model-based references under identical voltage correction. *Nested learning* [28] views a learning system as interconnected optimization problems, each updated at its own frequency; we use it as a design principle, with one level per timescale and with feasibility and voltage safety kept out of the learning problem.

The contributions of this paper are as follows.

1) **A nested three-timescale controller** (Section III): an hourly price-corridor policy trained with proximal policy optimization (PPO), a shared 15-min dispatch policy trained with deep deterministic policy gradient (DDPG) and a non-learned 60-s voltage layer, with the drivers' price response modeled as part of the environment rather than as a control level.
2) **Plan-residual dispatch that is feasible by construction.** Level 2 learns a correction to a deadline-aware cheapest-slot plan (residual policy learning [29], [30]), and least-laxity-first (LLF) allocation with a deadline guard maps it to per-vehicle rates within charger and transformer limits. A revenue-neutrality constraint on the retail price is enforced through a proportional–integral multiplier and a post-training tariff calibration applied to every learned method alike.
3) **Reactive voltage correction bounded by the inverter rating.** Level 3 uses $Q_i^{\max} = \sqrt{S_i^2 - P_i^2}$ [25], so curtailment never removes headroom; a secant step on the sensitivity measured between successive power flows closes the voltage gap, with curtailment as fallback. The layer is independent of the policy above it and is attached to every baseline, which makes the comparison of coordinators fair.
4) **A reproducible, paired evaluation** (Sections II and IV): AC power flow on two IEEE feeders; ENTSO-E prices [31], Pecan Street loads [32] and ACN-Data sessions [33]; 50 held-out days; five training seeds; eight baselines, including model-predictive control (MPC) with a linear-programming optimal power flow (LP-OPF) and perfect foresight, PPO-Lagrangian, constrained policy optimization (CPO) and hierarchical RL (HRL); every number, table and figure of the paper is regenerated by one command, and every qualitative statement is checked against the data.

The evaluation also reports where learning does *not* help: at nominal load a well-designed price-aware heuristic remains cheaper than the learned controller, and we attribute the controller's advantages under stress to its individual components.

Section II presents the system model and study design, Section III the controller, Section IV the experimental setup, Sections V and VI the results and their scope, and Section VII concludes.

II. SYSTEM MODEL AND STUDY DESIGN

We first fix the setting in which the controller acts: feeders, data, the drivers' price response and the metrics by which every method is judged.

A. Feeder and Power Flow

We use the IEEE 33-bus [34] and 69-bus [35] radial feeders with the MATPOWER data [36] (open tie lines removed). Five aggregators connect at electrically strong and weak buses (Table II); vehicles are assigned to them at random in proportions 0.18, 0.18, 0.19, 0.19 and 0.26, and aggregator k serves them through a transformer of rating $P_k^{\mathrm{cap}} = 500\,\mathrm{kW}$. With constant-power loads the radial AC power-flow equations are

$$\mathbf{V} = V_0\mathbf{1} - \mathbf{Z}^{\mathrm{path}}\left(\mathbf{S} \oslash \mathbf{V}\right)^*, \tag{1}$$

where $\mathbf{V}$ and $\mathbf{S}$ hold the complex voltages and loads of the non-substation buses, $V_0 = 1.03\,\mathrm{p.u.}$ is the substation voltage, $\mathbf{1}$ is the all-ones vector, $\oslash$ is element-wise division, $(\cdot)^*$ is element-wise conjugation and Z_{nm}^{path} is the impedance of the path shared by buses n and m from the substation. We solve (1) every $60\,\mathrm{s}$ by the backward/forward sweep [37] to $10^{-10}\,\mathrm{p.u.}$; our tests match pandapower's Newton–Raphson solution [38] to $10^{-8}\,\mathrm{p.u.}$.

B. Data, Fleets, Driver Response and Metrics

Day-ahead prices from the ENTSO-E Transparency Platform [31] and the 15-min consumption of 25 Pecan Street homes [32], both as distributed with EV2Gym [14], drive the simulation (Table II). The household profile, normalized by its 99.5th percentile and matched by day of year (so training and test days share it), scales the nominal feeder loads. The scale is the largest value (in steps of 0.01) for which the feeder *without* EVs stays at or above $0.955\,\mathrm{p.u.}$ over the whole year, so the voltage problems studied here are caused by EV charging. The main experiments use a synthetic residential fleet with the stated distributions of Table II (initial state of charge clipped to $[0.05, 0.9]$, target at least 0.05 above it, battery classes equally likely) and chargers of rating $\bar{c} = 11\,\mathrm{kW}$ and efficiency $\eta = 0.95$. For generalization we use real sessions from ACN-Data [33] at Caltech and JPL (bundled with SustainGym [15]):

TABLE II
SIMULATION SETUP, READ FROM THE CONFIGURATION FILE AND THE CODE

Item	Value
Feeders	IEEE 33-/69-bus, 12.66 kV, $V_0 = 1.03$ p.u., floor 0.95 p.u.
Aggregators	buses 6, 12, 18, 25, 30 (33-bus), 9, 21, 35, 49, 61 (69-bus); 500 kW each; 300 households
Base load	Pecan Street profile, scale 0.78 / 0.75, noise σ 0.02
Prices	ENTSO-E NL day-ahead; train 2023, test 2024, regime 2019
Arrival, dwell	$\mathcal{N}(18, 2^2)$ h in [12, 23.5]; $\mathcal{N}(12, 3^2)$ h in [2, 16]
SoC, battery	$\mathcal{N}(0.35, 0.15^2)$ to $\mathcal{N}(0.85, 0.1^2)$; 40/60/75/100 kWh
Charger	11 kW, η 0.95, S_i 12 kVA
Acceptance	λ^{ref} €0.20/kWh, w_i^{c} $\mathcal{N}(5, 1^2)$, w_i^{n} $\mathcal{N}(2, 0.5^2)$, b_i $\mathcal{N}(0, 0.5^2)$
Timing	power flow 60 s, dispatch 15 min, pricing 1 h; day from 12:00; look-ahead 12 h

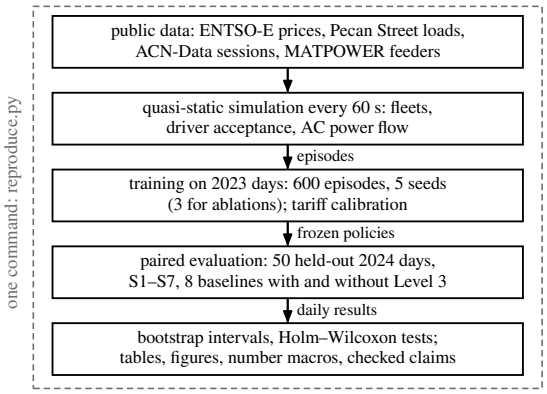


Fig. 1. Study pipeline from public data to every reported number; all stages run from one command.

each sampled vehicle keeps a recorded session's arrival, dwell and delivered energy (2019 for training, 2020–2021 for testing, weekdays, episodes from 06:00). Requests are capped at what the charger can deliver during the dwell.

Price response is part of the environment, not a control level. Each vehicle accepts the execution price p_k of its aggregator with the Fishbein–Ajzen attitude model [39],

$$P_i^{\mathrm{acc}} = \sigma\left(w_i^{\mathrm{c}} \frac{\lambda^{\mathrm{ref}} - p_k}{\lambda^{\mathrm{ref}}} + w_i^{\mathrm{n}} \bar{A} + b_i \right), \tag{2}$$

with logistic σ, reference rate λ^{ref}, last-interval fleet acceptance $\bar{A}$ (floored at 0.3) and the heterogeneous weights of Table II. A driver whose remaining need exceeds 0.7 times the charger rating times the hours to departure accepts regardless of price. Acceptance is drawn once per dispatch interval.

A vehicle draws power only while plugged in, having accepted the price and still needing energy. The energy cost is the day-ahead price times the energy drawn after any curtailment. We report the daily cost (€), the cost per delivered kWh, the service quality (SQ, delivered over requested energy), the violation rate (Viol., share of 60-s steps with a bus below $\underline{V} = 0.95$ p.u. after any correction), the mean daily minimum voltage (lowest bus voltage $V^{\min}$ of each day, averaged over days; $V^{\min}$ denotes the lowest bus voltage of the period considered), curtailed (Curt.) and unmet energy (need left at departure), both in kWh per day, and the mean retail price paid (Retail). Fig. 1 summarizes the study from data to results.

III. NESTED CONTROLLER

The controller splits coordination by timescale, so that each decision is taken at the rate at which its inputs change. Fig. 2 shows the architecture: each level is a decision process at its own rate whose environment includes the drivers' price response and the faster levels.

A. Level 1: Price Corridor (PPO, Hourly)

The state $s_t^{(1)} \in \mathbb{R}^{11}$ holds the mean bus voltage, substation P and Q, current price, plugged-in share, mean state of charge, line-loading margin (rating 0.4 kA), substation loading (8 MVA), the hour (sine, cosine) and the mean price over the next 12 h. The action $a_t^{(1)} \in [0,1]^2$, drawn from Beta distributions, sets the center c_t and width w_t of the corridor $[p_t^{\min}, p_t^{\max}] = [c_t - w_t/2,\, c_t + w_t/2]$, clipped to $[p^{\mathrm{lo}}, p^{\mathrm{hi}}] = [€0.05, €0.40]/\mathrm{kWh}$:

$$c_t = \lambda^{\mathrm{ref}} + (2a_{t,1}^{(1)} - 1) \begin{cases} \lambda^{\mathrm{ref}} - p^{\mathrm{lo}}, & a_{t,1}^{(1)} < 0.5, \\ p^{\mathrm{hi}} - \lambda^{\mathrm{ref}}, & \text{otherwise}, \end{cases}$$
$$w_t = w^{\min} + a_{t,2}^{(1)} \,(w^{\max} - w^{\min}), \tag{3}$$

with $w^{\min} = €0.02$ and $w^{\max} = €0.20/\mathrm{kWh}$. The neutral action $(0.5, 0.5)$ gives the *neutral corridor* $[€0.145, €0.255]/\mathrm{kWh}$ around the reference rate, so an untrained policy charges about the flat tariff of the baselines. The reward of hour t,

$$r_t^{(1)} = -\frac{C_t}{100} - 50\,(20\,\Delta V_t)^2 - 10\,\mathbb{I}[\Delta V_t > 0] - 5\,\frac{U_t}{100} - 0.5\,\frac{X_t}{100} + \omega\,\frac{R_t - \lambda^{\mathrm{ref}} E_t^{\mathrm{del}}}{100}, \tag{4}$$

penalizes energy cost C_t, the voltage deficit $\Delta V_t = \max(0, \underline{V} - V_t^{\min})$ ($\mathbb{I}$: indicator), unmet energy U_t of departing vehicles and curtailment X_t; the last term, with retail revenue R_t and delivered energy E_t^{del}, enforces revenue neutrality (below). PPO [40] trains the policy (clipping 0.2, $\gamma = 0.99$, generalized advantage estimation (GAE) with $\lambda_{\mathrm{GAE}} = 0.95$).

B. Level 2: Plan-Residual Dispatch (DDPG, 15 min)

State. The state $s_{\tau,k}^{(2)} \in \mathbb{R}^{28}$ of aggregator k holds its bus voltage, the hour, the prices of the current and next 11 hours, the corridor, the number of plugged-in vehicles, their remaining need (total and in four laxity bins: below 1, 1–3, 3–6 and above 6 h), the power they can absorb, their mean remaining dwell, the last acceptance rate and the plan set point $u_{\tau,k}^0$. One actor μ and one critic are shared by the five aggregators, with a one-hot identifier (parameter sharing [41]).

Planning prior. For each plugged-in vehicle i with remaining need e_i, the plan selects the $\lceil e_i/(\eta \bar{c} \Delta) \rceil$ cheapest intervals before departure ($\Delta = 15$ min; prices beyond the 12-h look-ahead are valued at their known mean). The plan set point $u_{\tau,k}^0 \in [0,1]$ is the power of the vehicles whose current interval is selected, as a fraction of the power the aggregator can deliver

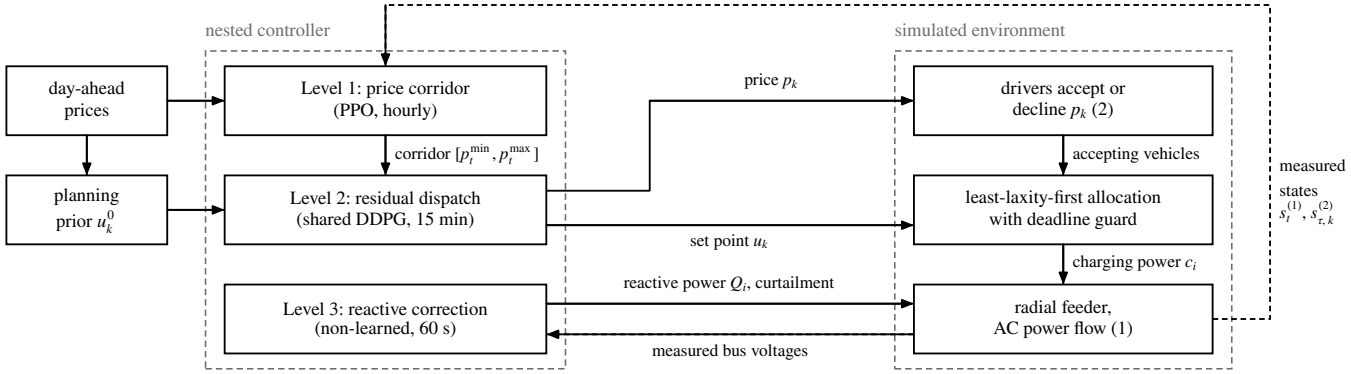


Fig. 2. Nested controller and simulated environment. Solid arrows are set points and data passed down; dashed arrows are measurements fed back. Labels name the quantity passed; numbers in parentheses refer to equations.

now; it uses only the price look-ahead and the needs and departures of plugged-in vehicles.

Action. The actor output $a_{\tau,k} = \mu(s^{(2)}_{\tau,k}) \in [0,1]^2$ corrects the plan and sets the execution price inside the corridor:

$$u_{\tau,k} = \mathrm{clip}\big(u^0_{\tau,k} + \rho\,(2a_{\tau,k,1} - 1),\, 0,\, 1\big), \tag{5}$$

$$p_{\tau,k} = p_t^{\min} + a_{\tau,k,2}\,(p_t^{\max} - p_t^{\min}), \tag{6}$$

so $a_{\tau,k,1} = 0.5$ reproduces the plan (residual scale ρ in Table III).

Feasibility by construction. Let $\hat{c}_i = \min\{\bar{c},\, e_i/(\eta\Delta)\}$ for the vehicles that accepted the price. The set point dispatches $P_{\tau,k} = u_{\tau,k}\min\{P_k^{\mathrm{cap}}, \sum_i \hat{c}_i\}$, filled in order of increasing laxity $\ell_i = h_i - e_i/(\eta\bar{c})$ (h_i: hours to departure) [7]; a deadline guard raises $P_{\tau,k}$ to cover every vehicle with $\ell_i \le \Delta$, up to the same limit. Every dispatch therefore lies in

$$\mathcal{C}_k = \Big\{c:\ 0 \le c_i \le \hat{c}_i,\ \textstyle\sum_i c_i \le P_k^{\mathrm{cap}}\Big\}. \tag{7}$$

Per-vehicle rates of other methods are projected onto $\mathcal{C}_k$.

Reward. The reward of aggregator k in interval τ is

$$\begin{aligned} r^{(2)}_{\tau,k} = &-C_{\tau,k} + \omega\,(p_{\tau,k} - \lambda^{\mathrm{ref}})E^{\mathrm{del}}_{\tau,k} \\ &- 2\,U_{\tau,k} - 0.5\,G_{\tau,k} - 0.5\,X_{\tau,k}, \end{aligned} \tag{8}$$

with the interval's energy cost $C_{\tau,k}$ (summing to the daily cost), unmet energy $U_{\tau,k}$ of departing vehicles, energy $G_{\tau,k}$ that can no longer be delivered before departure and curtailed energy $X_{\tau,k}$ (kWh; money in €). With 96 decisions per day we use $\gamma = 0.999$; $\gamma = 0.99$ would discount a cost 12 h ahead by almost 40%.

Revenue-neutral pricing. A profit-seeking aggregator would raise prices, lowering acceptance and pushing charging into expensive hours. We instead require the mean retail price paid to equal the flat rate λ^{ref} of the non-pricing baselines; ω in (4) and (8) is the multiplier of this constraint, updated after every episode by a proportional–integral (PI) rule [42] because plain dual ascent oscillated:

$$\varepsilon = (\lambda^{\mathrm{ref}} - \bar{p})/\lambda^{\mathrm{ref}}, \qquad \bar{\varepsilon} \leftarrow (1-\alpha)\,\bar{\varepsilon} + \alpha\,\varepsilon,$$

$$\omega = \mathrm{clip}\Big(K_p\,\bar{\varepsilon} + K_i \textstyle\sum \varepsilon,\ \omega^{\min},\ \omega^{\max}\Big), \tag{9}$$

where $\bar{p}$ is the episode's mean retail price paid, $\bar{\varepsilon}$ an exponential moving average (EMA) with weight α, and the sum over past episodes is clipped against wind-up (Table III). Since this holds the constraint only on average over training, a constant price offset (the sum clipped to $[p^{\mathrm{lo}}, p^{\mathrm{hi}}]$) is then calibrated by bisection on ten 2023 training days so that the deterministic policy's mean retail price equals λ^{ref}; this calibration is applied to every learned method and moved prices by at most €0.033/kWh.

TABLE III
HYPERPARAMETERS, READ FROM THE CONFIGURATION FILE AND THE CODE

Level 1 (PPO)	actor/critic 256–128–64, Beta policy, lr 0.0003, γ 0.99, λ_{GAE} 0.95, clip 0.2, 10 epochs per update, entropy 0.001
Level 2 (DDPG)	actor/critic 256–128–64, shared, one-hot id lr actor/critic 0.0001/0.001, γ 0.999, τ 0.005 batch 128, buffer 200 000, warm-up 1 000 transitions Gaussian action noise, std 0.3→0.05 (factor 0.995 per episode); residual scale ρ 0.1
Multiplier	K_p 2.0, K_i 0.05, EMA weight α 0.2, $\omega \in [-2.0, 2.0]$, $\lvert K_i \sum e\rvert \le 2.0$
Training	600 episodes; curriculum window 20
Baselines	same networks, learning rates, episodes and curriculum; safe RL: γ 0.999, cost limit 0, dual step 0.05 (PPO-Lag.), KL radius 0.01 (CPO)
Level 3	δ 0.001 p.u., S_i 12.0 kVA, ≤8 (+12 when curtailing) power flows

Training. DDPG [43] trains actor and critic (256–128–64 units) on the action *actually applied* after the guard, mapped back through (5) and clipped to $[0,1]$; the actor's output layer starts near zero, so training starts from the plan. Episodes are random 2023 days. A curriculum raises the fleet from 60 to 300 vehicles in steps of 60 when the last 20 episodes reach mean SQ ≥ 0.95 with mean curtailment $\le 20\,\mathrm{kWh}$; the final stage draws 120 to 300 vehicles. Each run has 600 episodes (about 0.5 h on one CPU core); Table III lists the hyperparameters of both learned levels.

C. Level 3: Reactive Voltage Correction (60 s)

Level 3 runs inside every 60-s step (Fig. 3). If $\min_n |V_n| < \underline{V} + \delta$ after the power flow ($\delta = 0.001\,\mathrm{p.u.}$), the plugged-in

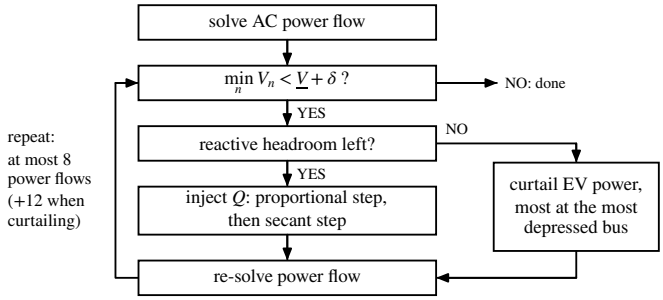


Fig. 3. Level-3 correction inside one 60-s step.

chargers at aggregator bus k offer the headroom

$$\bar{Q}_k = \sum_{i \in k} \sqrt{\max\{S_i^2 - P_i^2,\, 0\}}, \qquad S_i = 12\,\text{kVA}, \quad (10)$$

so a charger still offers 4.8 kvar at its full 11 kW and its full rating at $P_i = 0$, and curtailing active power *increases* the headroom (S_i bounds only the reactive capability; the 11-kW active-power limit is separate). The first injection is proportional to the voltage gap, saturates at $V^{\min} = 0.90\,\text{p.u.}$ and is weighted toward the most depressed aggregator buses. Later iterations measure the sensitivity $\hat{s} = \Delta V^{\min}/\Delta Q$ between the last two power flows and, if ΔQ and $\Delta V^{\min}$ are both positive, take the secant step

$$\Delta Q = 1.2\left(\underline{V} + \delta - V^{\min}\right)/\hat{s}, \quad (11)$$

split by voltage depression and capped by the remaining headroom; otherwise the proportional step is repeated. Without headroom, EV power is shed in steps of up to 20% per iteration, most at the most depressed bus. The loop stops when $V^{\min} \geq \underline{V} + \delta$ or after eight corrective power flows (twenty once curtailment has started). Measured on successive AC solutions, the step needs no precomputed network model. Level 3 is independent of the policy above it, so we also attach it to every baseline ("+L3").

IV. Experimental Setup

A fair test needs strong alternatives, stressed conditions and many independent days. Table IV defines the scenarios. S1, S2, S3 and S5 vary the penetration. S4 repeats S3 with an hourly-persistent deviation of the base load from its profile (standard deviation 15%, clipped to $\pm 40\%$), a forecast error for the LP-OPF, the only method that uses a load forecast. Two stress tests probe Level 3: S6 forbids charging between 19:00 and 20:00, so that charging resumes in a synchronized rebound, and S7 emulates reduced reactive capability by halving the inverter rating to $S_i = 6\,\text{kVA}$, so a charger drawing more than 6 kW offers no reactive power.

The nested controller is compared with eight baselines. *Rule-based:* uncoordinated charging (full power on arrival); a TOU timer (off-peak 23:00–07:00, earlier only when laxity is exhausted; tariff €0.12/kWh off-peak, €0.30/kWh 17:00–21:00 and €0.20/kWh otherwise); and the price-aware heuristic, i.e., the Level-2 plan executed per vehicle at full power with the same 12-h look-ahead. *Model-predictive LP-OPF*:*

TABLE IV
Evaluation Scenarios (Penetration: EVs per Household)

Scenario	EVs	Pen. (%)	Condition
S1	120	40	nominal
S2	200	67	nominal
S3	240	80	nominal (main case)
S4	240	80	base load $\pm 15\%$ hourly
S5	300	100	nominal
S6	240	80	no charging 19:00–20:00
S7	360	120	inverter rating $\times 0.5$

at every interval an LP over the rest of the day (HiGHS [44]) minimizes energy cost plus €5/kWh of unmet energy subject to energy, charger and transformer limits and linearized voltage limits ($\geq \underline{V} + \delta$) from finite-difference sensitivities at the forecast no-EV operating point. The asterisk marks *perfect foresight* of prices and of every vehicle's arrival, departure and need. It is a strong reference, neither deployable nor a strict bound: its voltage model is linearized and drivers may decline. *Learned:* flat DDPG (one agent for all aggregators), PPO-Lagrangian and CPO [18] (expected-cost constraint on violations, limit $d = 0$) and a goal-conditioned HRL agent (hourly DDPG goals, 15-min DDPG workers). They observe the same states, use the same allocation, reward terms, discount and revenue-neutral multiplier, and price inside the neutral corridor; flat DDPG and HRL add a voltage penalty (HRL workers also a goal-tracking term). All are evaluated with and without Level 3. The non-pricing baselines charge λ^{ref}.

All methods are evaluated on the same 50 distinct 2024 days with identical fleets and noise (paired design); learned methods use five training seeds (three in ablations and generalization), averaged per day. We report means with 95% bootstrap confidence intervals over days (10 000 resamples) and Holm-corrected Wilcoxon signed-rank tests on the paired daily differences. The allocation rule was fixed before testing on 30 held-out 2023 days by a pre-specified criterion: over three variants that serve planned slots first (two seeds each), the S3 cost fell by up to 1.3%, but 24%–89% (S3) and 39%–103% (S7) more energy was left undelivered, so least-laxity-first allocation was kept.

V. Results

We compare all methods in the main case and on the grid side, then vary the penetration and test significance, then stress the controller and dissect it, and finally test its transfer to other feeders, fleets and years.

A. Main Comparison and Grid Impact

The main case, 80% penetration on the 33-bus feeder (S3), is compared in Table V. Uncoordinated charging costs €931.4 per day and violates the voltage floor in 3.99% of the 60-s steps. Attaching Level 3 removes every violation of the rule-based methods at S1, S2, S3 and S5 and changes their cost by at most 0.10%. The perfect-foresight LP-OPF is cheapest (€612.0), followed by the price-aware heuristic, the cheapest method without foresight (€619.8). The nested controller costs

TABLE V
RESULTS AT S3 (240 VEHICLES, 80% PENETRATION; IEEE 33-BUS; 50 HELD-OUT DAYS; LEARNED METHODS AVERAGED OVER FIVE TRAINING SEEDS). COST: MEAN DAILY COST WITH 95% BOOTSTRAP CONFIDENCE INTERVAL; COST/KWH AND RETAIL IN €/KWH; OTHER METRICS AS DEFINED IN SECTION II-B; +L3: LEVEL 3 ATTACHED

Method	Cost (€)	Cost/kWh	Retail	SQ	Viol. (%)	Mean $V^{\min}$	Curt. (kWh)
Uncoordinated	931.4 [833.8, 1060.7]	0.1146	0.200	1.000	3.99	0.9554	0.0
Uncoordinated + L3	931.4 [835.0, 1062.6]	0.1146	0.200	1.000	0.00	0.9615	5.0
TOU timer	697.7 [639.6, 751.2]	0.0863	0.132	0.995	4.30	0.9372	0.0
TOU timer + L3	697.1 [641.0, 750.9]	0.0862	0.132	0.995	0.00	0.9515	23.7
Price-aware heuristic	619.8 [561.8, 674.2]	0.0766	0.200	0.997	0.73	0.9598	0.0
Price-aware heuristic + L3	619.8 [564.4, 675.0]	0.0766	0.200	0.997	0.00	0.9623	1.7
MPC LP-OPF (perfect foresight)	612.0 [555.7, 667.2]	0.0759	0.200	0.992	0.01	0.9634	0.0
MPC LP-OPF (perfect foresight) + L3	612.0 [555.3, 667.3]	0.0759	0.200	0.992	0.00	0.9635	0.0
Flat DDPG	756.1 [695.5, 828.3]	0.0934	0.203	0.996	0.45	0.9673	0.0
Flat DDPG + L3	756.1 [694.6, 827.0]	0.0934	0.203	0.996	0.00	0.9686	0.1
PPO-Lagrangian	790.2 [725.8, 866.6]	0.0976	0.199	0.996	0.19	0.9758	0.0
PPO-Lagrangian + L3	790.2 [725.2, 865.0]	0.0976	0.199	0.996	0.00	0.9763	0.0
CPO	816.9 [747.5, 900.1]	0.1006	0.200	0.999	0.35	0.9731	0.0
CPO + L3	817.0 [748.1, 903.3]	0.1006	0.200	0.999	0.00	0.9739	0.0
Hierarchical RL	741.3 [685.3, 799.4]	0.0915	0.200	0.997	0.03	0.9741	0.0
Hierarchical RL + L3	741.3 [685.3, 800.0]	0.0915	0.200	0.997	0.00	0.9743	0.0
Nested (proposed)	645.0 [590.0, 700.0]	0.0797	0.201	0.997	0.00	0.9665	0.4

TABLE VI
GRID SIDE OF S3: MEAN DAILY PEAKS OF EV AND FEEDER LOAD, SHARE OF 60-S STEPS WITH LEVEL 3 ACTIVE, REACTIVE ENERGY, CURTAILMENT AND POWER FLOWS PER DAY

Method	Peak EV (kW)	Peak feeder (MW)	L3 act. (%)	Q energy (kvarh)	Curt. (kWh)	PF per day
Uncoord.	1340	3.17	0.00	0.0	0.0	1440
Uncoord.+L3	1342	3.16	4.33	303.7	5.0	1567
TOU+L3	2305	3.65	4.59	531.6	23.7	1616
Price-aware+L3	1804	2.88	0.83	42.6	1.7	1467
LP-OPF*+L3	1715	2.79	0.05	0.5	0.0	1442
HRL+L3	1024	2.43	0.05	0.9	0.0	1442
Nested	1642	2.69	0.32	14.9	0.4	1449

€645.0, which is 30.8% less than uncoordinated charging and 4.1% more than the price-aware heuristic, with no violations and a retail price of €0.201/kWh. It is the cheapest learned controller, 13.0% below the best learned baseline (Hierarchical RL + L3).

For a distribution operator, peaks and corrective effort matter as much as cost; Table VI views S3 from this side. The nested controller lowers the mean daily peak of the feeder load by 15% relative to uncoordinated charging and by 7% relative to the price-aware heuristic; the hierarchical agent has the lowest peak, at a higher energy cost. The TOU timer starts most charging when its off-peak window opens and has the highest EV peak. Level 3 is cheap to run: it adds 0.7% to the 1440 power flows per day of the nested controller and at most 12.3% for any method (the TOU timer). Fig. 4 shows one representative test day. Uncoordinated charging starts on arrival and peaks at a higher price than the nested controller, and drops the lowest bus voltage to 0.938 p.u. With Level 3 attached, injections of up to 276 kvar in 8.9% of the steps keep it at 0.951 p.u. without curtailment. The heuristic and the nested controller move charging into the night price valley and stay above the floor without any reactive injection; their energy costs that day are €434.3 and €460.0, against €859.4.

B. Penetration, Base-Load Deviation and Statistical Comparison

Whether these results hold as more vehicles connect is shown by Table VII and Fig. 5, which extend the comparison to S1–S5. Without Level 3, violations of uncoordinated and TOU charging grow with penetration, up to 6.82% and 7.55%. Without base-load deviation (S1, S2, S3, S5) the nested controller and every "+L3" method record no violations. In S4 the household load alone breaks the floor on 4 of the 50 days; Level 3 removes most of these violations with the headroom of idle chargers, and the remainder (at most 0.005% of the steps for the nested controller) occurs only on those days. At every penetration level the nested controller costs 30.6%–31.0% less than uncoordinated charging and 3.7%–4.3% more than the price-aware heuristic, and it stays below every learned baseline by at least 12.6%.

Differences of a few percent need a significance test. Table VIII compares like with like: every baseline has Level 3 attached. Without it, the mean differences change by at most €0.7 and the same cost differences are significant. The cost differences to uncoordinated charging and to every learned baseline are significant after Holm correction, and so is the cost premium relative to the price-aware heuristic and the LP-OPF. Service-quality differences are at most 0.005 in absolute terms; the nested controller's service quality is significantly lower only than that of uncoordinated charging and CPO. Table IX adds the spread over training seeds: the standard deviation of the nested controller's mean daily cost is €3.1, against €11–54 for the learned baselines.

C. Stress Tests and Contribution of the Learned Levels

Two stress tests probe the limits of Level 3. In S6, after an hour without charging in the evening peak, every controller with Level 3 in Table X keeps the floor. S7 exceeds what the reduced reactive capability can correct: all methods except the perfect-foresight LP-OPF keep some violations. The

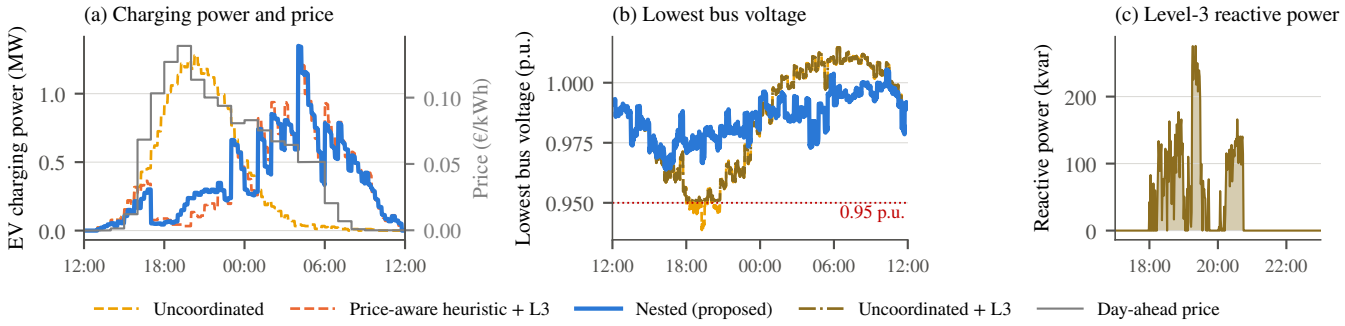


Fig. 4. Operation on a representative S3 test day (29 June 2024, the day with the median cost of uncoordinated charging; nested controller of training seed 0) at 60-s resolution. (a) Aggregate EV charging power and day-ahead price, (b) lowest bus voltage (uncoordinated charging with and without Level 3 coincide except where Level 3 acts), and (c) reactive power injected by Level 3 under uncoordinated charging; the heuristic and the nested controller need none on this day.

TABLE VII
DAILY ENERGY COST (€) AND VIOLATION RATE (%) IN S1–S5 (PENETRATION IN PARENTHESES; S4: S3 WITH BASE-LOAD DEVIATION). THE LEARNED BASELINES WITH LEVEL 3 (NOT SHOWN) HAVE NO VIOLATIONS IN S1, S2, S3 AND S5

Method	S1 (40%)		S2 (67%)		S3 (80%)		S4 (80%)		S5 (100%)	
	Cost	Viol.	Cost	Viol.	Cost	Viol.	Cost	Viol.	Cost	Viol.
Uncoordinated	474	0.42	780	2.35	931	3.99	931	4.53	1161	6.82
TOU timer	358	0.06	587	2.39	698	4.30	695	4.31	858	7.55
Uncoordinated + L3	474	0.00	780	0.00	931	0.00	931	0.02	1161	0.00
TOU timer + L3	358	0.00	587	0.00	697	0.00	695	0.00	858	0.00
Price-aware heuristic + L3	317	0.00	520	0.00	620	0.00	618	0.02	773	0.00
MPC LP-OPF (perfect foresight) + L3	313	0.00	513	0.00	612	0.00	610	0.00	765	0.00
Flat DDPG	409	0.09	642	0.30	756	0.45	754	0.84	931	1.11
PPO-Lagrangian	403	0.01	662	0.16	790	0.19	788	0.62	984	0.51
CPO	420	0.03	684	0.27	817	0.35	814	0.93	1016	0.81
Hierarchical RL	380	0.00	622	0.04	741	0.03	739	0.38	922	0.19
Nested (proposed)	329	0.00	540	0.00	645	0.00	642	0.00	806	0.00

TABLE VIII
PAIRED COMPARISON AT S3 WITH THE BASELINES WITH LEVEL 3 (Δ: NESTED MINUS BASELINE, 95% BOOTSTRAP INTERVAL; DAYS: DAYS OF 50 ON WHICH NESTED IS CHEAPER; WILCOXON TESTS, HOLM OVER 16 COMPARISONS PER METRIC)

Nested vs.	ΔCost (€)	p_{Holm}	Days	ΔSQ	p_{Holm}
Uncoord.+L3	−286.4 [−377.8, −219.2]	<0.001	50	−0.0028	<0.001
TOU+L3	−52.0 [−71.9, −34.0]	<0.001	41	+0.0016	0.011
Price-aware+L3	+25.2 [+21.9, +28.8]	<0.001	0	+0.0001	1.000
LP-OPF*+L3	+33.0 [+29.0, +37.3]	<0.001	0	+0.0046	<0.001
Flat DDPG+L3	−111.0 [−141.9, −89.8]	<0.001	50	+0.0009	<0.001
PPO-Lag.+L3	−145.2 [−179.5, −118.8]	<0.001	50	+0.0011	<0.001
CPO+L3	−171.9 [−216.2, −138.7]	<0.001	50	−0.0018	<0.001
HRL+L3	−96.2 [−108.5, −85.5]	<0.001	50	−0.0001	1.000

TABLE IX
LEARNED METHODS AT S3: MEAN DAILY COST, ITS STANDARD DEVIATION OVER FIVE TRAINING SEEDS, SERVICE QUALITY AND TRAINING TIME PER SEED (ONE CPU CORE)

Method	Cost (€)	Seed s.d. (€)	SQ	Training (h)
Flat DDPG	756.1	53.7	0.996	0.14
PPO-Lagrangian	790.2	43.7	0.996	0.06
CPO	816.9	10.9	0.999	0.07
Hierarchical RL	741.3	24.5	0.997	0.45
Nested (proposed)	645.0	3.1	0.997	0.47

nested controller curtails 67% less energy than the price-aware heuristic, leaves 64% less energy undelivered and has 92% fewer violations; its service quality is 0.997 against 0.991. The price is energy cost: €982.9 against €940.4 per day.

To see where these differences come from, Table XI separates the learned levels (Level 3 active). Executed without learning, the plan costs 1.3% more at S3 than the per-vehicle price-aware heuristic, because the aggregate set point is filled least-laxity-first rather than by each vehicle's own cheapest slots. Relative to the plan, learned prices alone raise the S3 cost by 1.0%, learned dispatch alone by 2.8%, and the full controller by 2.7%: at nominal load, learning does not lower the cost. Under stress, the lower unmet energy needs no learning: the plan executed least-laxity-first leaves 65% less energy undelivered than the per-vehicle heuristic in S7, and the full controller leaves about as much undelivered as the plan (38.2 against 37.0 kWh per day). The lower curtailment comes from learning: the full controller curtails 60% less energy than the plan, learned dispatch alone 38% less and learned prices alone 28% less.

D. Ablations, Generalization and Level-3 Parameters

Each component is then removed in turn (Table XII); a cost difference is called resolved when the per-seed means (three seeds) do not overlap with those of the complete framework.

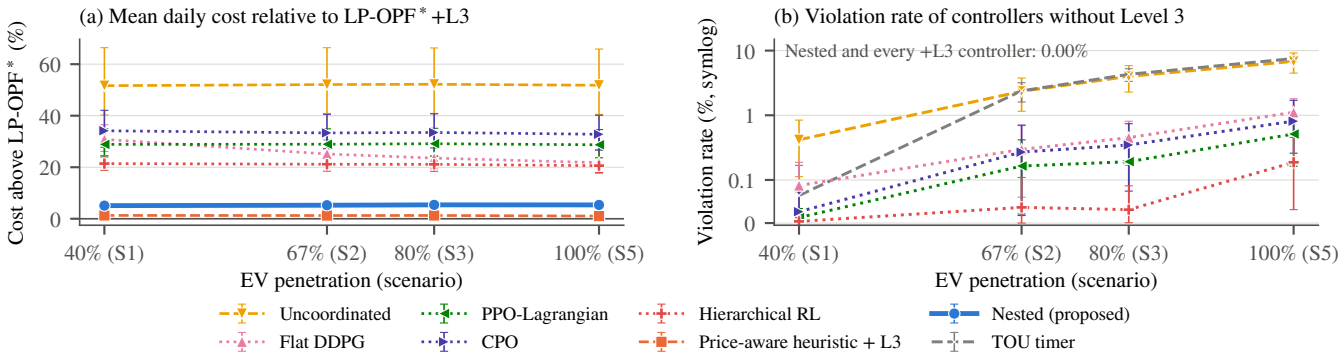


Fig. 5. (a) Mean daily energy cost relative to that of the perfect-foresight LP-OPF with Level 3 (ratio of means over the test days; 95% intervals from a bootstrap over days that keeps both methods' costs of a day together), and (b) violation rate of the controllers without Level 3 (symmetric-log axis), across S1, S2, S3 and S5. The nested controller and every "+L3" controller have no violations in these scenarios.

TABLE X
LEVEL-3 STRESS TESTS (S6: NO CHARGING 19:00–20:00; S7: 360 VEHICLES, $S_i = 6$ KVA). VIOL. IN %; $V^{\min}$: MEAN DAILY MINIMUM VOLTAGE (P.U.); CURT. AND UNMET IN KWH PER DAY

Method	S6		S7			
	Viol.	$V^{\min}$	Viol.	Curt.	Unmet	SQ
Uncoord.	3.80	0.951	9.00	0	6	1.000
TOU	4.32	0.937	10.62	0	402	0.968
Uncoord.+L3	0.00	0.958	0.35	492	15	0.999
TOU+L3	0.00	0.951	0.17	682	561	0.955
Price-aware+L3	0.00	0.962	0.16	139	106	0.991
LP-OPF*+L3	0.00	0.963	0.00	0	98	0.992
Nested	0.00	0.966	0.01	46	38	0.997

TABLE XI
CONTRIBUTION OF THE LEARNED LEVELS (LEVEL 3 ACTIVE). PLAN: PLANNING PRIOR WITHOUT LEARNING; PRICES ONLY: LEARNED PRICES, PLAN DISPATCH; DISPATCH ONLY: LEARNED DISPATCH, FLAT TARIFF. UNITS AS IN TABLE X

Configuration	S3			S7			
	Cost	Retail	SQ	Cost	Viol.	Curt.	Unmet
Price-aware+L3	619.8	0.200	0.997	940.4	0.16	139	106
Plan+L3	627.9	0.200	0.997	956.7	0.02	114	37
Prices only	633.9	0.201	0.997	966.3	0.03	82	43
Dispatch only	645.3	0.200	0.998	981.4	0.03	71	30
Nested	645.0	0.201	0.997	982.9	0.01	46	38

TABLE XII
ABLATIONS AT S3 AND S5, THREE TRAINING SEEDS EACH (COMPLETE FRAMEWORK ON THE SAME SEEDS). COST IN €, VIOL. IN %; –: NOT EVALUATED

Configuration	S3			S5		
	Cost	SQ	Viol.	Cost	SQ	Viol.
Complete framework	645.7	0.997	0.00	806.6	0.997	0.00
No planning prior	696.5	0.996	0.00	865.5	0.996	0.00
No Level 1	642.8	0.996	0.00	804.6	0.996	0.00
Single timescale	651.2	0.996	0.00	812.8	0.996	0.00
No behavior model	639.4	1.000	0.00	800.0	0.999	0.00
No Level 3 (trained)	648.5	0.996	0.23	810.5	0.995	0.71
Level 3 removed at test	645.7	0.997	0.33	–	–	–
No curriculum	639.2	0.996	0.00	797.3	0.996	0.00
No deadline guard	651.2	0.996	0.00	814.6	0.996	0.00
Proportional allocation	660.8	0.995	0.00	828.5	0.995	0.00

TABLE XIII
GENERALIZATION AT 80% PENETRATION (THREE TRAINING SEEDS). UNC.: UNCOORDINATED; PA: PRICE-AWARE HEURISTIC; LP*: PERFECT-FORESIGHT LP-OPF

Setting	Daily cost (€)				Viol. (%)		SQ
	Unc.	PA+L3	LP*+L3	Nested	Unc.	Nested	Nested
33-bus, residential	931.4	619.8	612.0	645.0	3.99	0.00	0.997
69-bus, residential	931.4	619.8	611.7	648.0	0.21	0.00	0.997
33-bus, ACN Caltech	146.5	121.3	119.7	126.0	0.00	0.00	0.970
33-bus, ACN JPL	263.8	194.1	197.1	207.0	0.00	0.00	0.994
33-bus, 2019 days	371.1	282.3	280.2	289.9	4.72	0.00	0.997

The planning prior is resolved at S3 and S5: learning the dispatch from scratch raises the cost by 7.9% and 7.3%. Removing the deadline guard is resolved only at S5 (+1.0%). Proportional allocation, a single timescale (Level 1 acting every 15 min) and a fixed neutral corridor instead of Level 1 change the cost by at most 2.8%, within the seed spread; without Level 1 the S3 cost is even 0.5% lower, so we attribute no cost benefit to the hourly price level. Without the curriculum the cost is lower by 1.0% (S3) and 1.2% (S5), a resolved difference. Without the behavioral model every driver accepts, which raises service quality and lowers the cost by 1.0%: price response constrains the controller. A policy trained without Level 3 violates the floor in 0.23% of the steps, and removing Level 3 from the trained controller at test time gives 0.33%.

To test transfer beyond the training setting, Table XIII repeats S3 on the 69-bus feeder, on recorded sessions from two workplace sites and on 2019 days (with the policies trained on 2023 days). In every setting the nested controller costs 14.0%–30.8% less than uncoordinated charging and 2.7%–6.6% more than the price-aware heuristic, with no violations. At JPL the price-aware heuristic is cheaper than the LP-OPF, a reference, not a bound. At the workplace sites even uncoordinated charging has no violations at Caltech and violations in at most 0.005% of the steps at JPL. At Caltech every method, uncoordinated charging included, delivers a smaller share of the requested energy than in any other setting (service quality

at most 0.971).

Finally, Level 3 has two parameters. Varying the correction margin ($\delta = 0.0005$, 0.001 and $0.002\,\mathrm{p.u.}$) and the inverter rating ($S_i = 10$, 12 and $14\,\mathrm{kVA}$, where $10\,\mathrm{kVA}$ emulates chargers with reduced reactive capability) at S3 causes no violations and changes the cost of the nested controller by at most 0.01%. A larger margin activates Level 3 in more of the 60-s steps (0.30%, 0.32% and 0.37%), and a smaller rating shifts the correction from reactive power to curtailment (0.0, 0.4 and 3.9 kWh per day at 14, 12 and $10\,\mathrm{kVA}$).

VI. Discussion

Some conclusions are firmer than others; we discuss what learning contributes, where the voltage correction applies and what limits the findings.

At nominal load, learning does not beat a well-designed rule on cost: the price-aware heuristic is cheaper than the nested controller, and neither the learned prices nor the learned dispatch reduce cost relative to the plan (Table XI). The learned levels reduce curtailment, which mattered under stress (S7); the lower unmet energy under stress comes from least-laxity-first allocation, not from learning. Learned dispatch needs the plan as a starting point: learned from scratch, it is costlier at both penetration levels tested (S3 and S5). The nested design is therefore not a way to beat optimization on cost but a way to keep learned components where they help while planning and physics handle the rest. We state this plainly because it is the finding most likely to be misread: the value of the controller lies in keeping the floor at every penetration tested, in its small spread over training seeds and in its behavior under stress, not in a lower energy bill than the heuristic.

Level 3 is corrective control on the simulated quasi-static state, not an a priori guarantee: it restores the floor whenever the chargers' reactive headroom suffices and otherwise sheds EV power. Three points qualify it. First, the inverter rating sets the voltage margin (Section V-D, S7). Second, we assume, as in [25], that a charger supplies reactive power at any active power, including zero. Third, the floor is applied at the medium-voltage ($12.66\,\mathrm{kV}$) buses; service-transformer and secondary drops, switching transients and sub-cycle dynamics are outside the scope, as in any quasi-static study.

The remaining assumptions bound the generality of the results. *Information:* the plan, the learned agents and the heuristic know each plugged-in vehicle's departure and need, as in managed-charging programs where drivers enter them [33], but not future arrivals. *Network:* the feeders are balanced, radial and single-phase equivalent; unbalance can change the sensitivities the correction uses. *Behavior:* the acceptance model (2) is stylized; the voltage layer does not depend on it, the value of price signals does. *Data:* the residential fleet is synthetic; loads (United States) and prices (Netherlands) come from different systems; training and test days share the base-load profile; market change is tested only through 2019.

VII. Conclusion

We presented a nested EV charging controller with hourly pricing, 15-min plan-residual dispatch and 60-s reactive voltage correction. On two IEEE feeders with real price, load and session data, it kept the voltage floor without base-load deviation, up to 100% penetration on the 33-bus feeder and at 80% in the other settings, and was the cheapest learned controller. A price-aware heuristic remained cheaper at nominal load, and a perfect-foresight LP-OPF cheaper still. Under stress beyond the chargers' reactive capability, the controller curtailed and left undelivered far less energy than the heuristic; the lower curtailment came from the learned levels, the lower unmet energy from least-laxity-first allocation. Two lessons follow. First, a physics-based correction layer bounded by the inverter rating removed the violations of every coordinator we attached it to, up to 100% penetration without base-load deviation, at an energy-cost change of at most 0.10%; it belongs in every comparison. Second, learned dispatch pays off when it starts from a sound plan and grid stress binds.

Several directions follow from the limitations stated above. The feeder model should be extended to unbalanced three-phase networks with service transformers and secondary circuits, where the voltage floor is usually binding first and where phase unbalance changes the sensitivities that Level 3 measures. The information assumption should be relaxed by learning under uncertain departures and needs, for instance from driver-entered estimates with recorded error distributions, and by forecasting arrivals. The stylized acceptance model should be replaced or calibrated with recorded driver responses to dynamic tariffs, since the value of the price level depends on it. Because learning did not lower the energy cost at nominal load, a lighter design that keeps the plan, the least-laxity-first allocation and Level 3 but drops the learned levels deserves a direct comparison on cost, robustness and implementation effort. Finally, vehicle-to-grid active power and the interaction with on-load tap changers and capacitor banks widen the set of corrective actions, and a hardware-in-the-loop or field trial is needed before any run-time claim about the voltage layer can be made for a real feeder.

Data and Code Availability

Code, configuration, trained policies, evaluation logs and the script that regenerates every table, figure and number are available at https://github.com/Abraheem13/nested-ev-grid. All data are openly available through the EV2Gym and SustainGym packages and the MATPOWER feeder files (Section II) and are verified by SHA-256 checksums. Household load data were provided by Pecan Street Inc. (Dataport).

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